CEFTRIAXONE

Never with calcium IV in neonates

Once daily

CHEMICAL NAME (6R,7R)-7-[[(2Z)-2-(2-amino-1,3-thiazol-4-yl)-2-methoxyiminoacetyl]amino]-3-[(2-methyl-5,6-dioxo-1H-1,2,4-triazin-3-yl)sulfanylmethyl]-8-oxo-5-thia-1-azabicyclo[4.2.0]oct-2-ene-2-carboxylic acid (disodium)

CLASS Third-generation cephalosporin (parenteral)

TRADE NAMES Rocephin · Triaxone · Cefaxone · Oframax

THERAPEUTIC USES / INDICATIONS

- Bacterial meningitis (empirical).
- Community-acquired pneumonia, sepsis, pyelonephritis.
- Gonorrhoea (single IM dose); typhoid fever; Lyme disease (late).
- Intra-abdominal infection (with metronidazole); surgical prophylaxis.

ADMINISTRATION

- **IV:** injection over 5 min or infusion over 30 min (60 min in neonates).
- **IM:** deep gluteal; may be reconstituted with 1% lidocaine (never give lidocaine solution IV).
- Do not mix or co-infuse with calcium-containing solutions (Ringer's, Hartmann's, TPN) — precipitate.

DOSAGES

- **Adult:** 1–2 g once daily (max 4 g/day).
- **Meningitis:** 2 g every 12 h.
- **Gonorrhoea:** 500 mg–1 g IM single dose.
- **Children:** 50–80 mg/kg once daily; meningitis 100 mg/kg/day (max 4 g).
- No adjustment in renal impairment alone.

PHARMACOKINETICS

- **Onset:** immediate (IV); peak IM 2–3 h.
- Highly protein-bound; good CSF penetration.
- Dual excretion: urine (~60%) and bile (~40%). **t½:** 6–9 h (allows once-daily dosing).

PHARMACODYNAMICS

- Inhibits **cell-wall synthesis** → bactericidal.
- Broad Gram-negative cover (*N. meningitidis*, *N. gonorrhoeae*, *H. influenzae*, Enterobacterales) and pneumococci; not *Pseudomonas*, enterococci, MRSA.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Diarrhoea, nausea.
- Rash; injection-site pain.
- Raised liver enzymes.

ADVERSE EFFECTS (SERIOUS)

- Anaphylaxis.
- Biliary sludge / pseudolithiasis (high doses).
- *C. difficile* colitis; haemolytic anaemia.
- Fatal calcium–ceftriaxone precipitates in lungs/kidneys of neonates.

CONTRAINDICATIONS & CAUTIONS

- Cephalosporin hypersensitivity; severe penicillin allergy.
- Neonates (≤28 days) needing IV calcium; premature neonates; jaundiced neonates (displaces bilirubin → kernicterus).
- Lidocaine-reconstituted solution must not be given IV.

KEY INTERACTIONS

- **Calcium-containing IV solutions:** precipitation.
- **Warfarin:** ↑ INR.

NURSING MANAGEMENT

- Check allergy history.
- Flush line with saline before and after; no calcium solutions in same line.
- Check reconstitution fluid: lidocaine for IM only.
- Monitor temperature, cultures, LFTs in long courses.
- Neonates: check bilirubin and calcium therapy.

PATIENT EDUCATION

- IM injection may be painful.
- Report rash, breathing difficulty, severe diarrhoea or right upper abdominal pain.

KEY POINTS

- 3rd-gen IV/IM cephalosporin; once daily.
- No calcium IV in same line (neonates fatal).
- Meningitis, sepsis, gonorrhoea, typhoid.
- Biliary + renal excretion.

SOURCES CHECKED **[135]** Ceftriaxone 1 g Powder for Solution for Injection or Infusion (UK SmPC) **[136]** Ceftriaxone, IM, IV (US label) **[137]** ceftriaxone, IM, IV (US label)

Full references p. 432–457